SIMULATION VARIABLES

SimVars are "simulation variables" and they are the way that third party APIs and add-ons can get information about the current state of the simulation. In general these variables are used to control aircraft behaviors by permitting you to retrieve data about things like weather conditions, position, or aircraft status, and in many cases they can also be used to modify something within the simulation, in particular aircraft instruments and controls - although they can also have many other purposes within any add-on package.

You can find detailed information about each particular SimVar on the following pages:

- Simulation Variable Units
- Camera Variables
- Services Variables
- Miscellaneous Variables
- Aircraft SimVars:
    - Aircraft Autopilot/Assistant Variables
    - Aircraft Brake/Landing Gear Variables
    - Aircraft Control Variables
    - Aircraft Electrics Variables
    - Aircraft Engine Variables
    - Aircraft Flight Model Variables
    - Aircraft Fuel Variables
    - Aircraft Misc. Variables

- Aircraft Radio Navigation Variables
- Aircraft System Variables
- Helicopter Variables

There is also a further section dedicated to **Event IDs**. These are used as a way to interface between different simulation systems, with the user at the top level the various sub-systems within the simulation at the bottom. You can find all these event IDs in the following section:

- Event IDs

Finally, there is an additional document for **Audio Design** that lists the RTPC events and the SimVars they are associated with:

- RTPC And Simulation Variables

SimVar Index

Below you can find indexed all the SimVars available, presented this way to facilitate finding the required variable at a glance or through the search system:

A

ACCELERATION BODY X
ACCELERATION BODY Y
ACCELERATION BODY Z
ACCELERATION WORLD X
ACCELERATION WORLD Y
ACCELERATION WORLD Z
ADF ACTIVE FREQUENCY
ADF AVAILABLE
ADF CARD
ADF EXT FREQUENCY ***Deprecated***
ADF FREQUENCY ***Deprecated***
ADF IDENT
ARTIFICIAL GROUND ELEVATION
ASSISTANCE LANDING ENABLED
ASSISTANCE TAKEOFF ENABLED
ATC AIRLINE
ATC AIRPORT IS TOWERED
ATC CLEARED IFR
ATC CLEARED LANDING
ATC CLEARED TAKEOFF
ATC CLEARED TAXI
ATC CURRENT WAYPOINT
AUTOPILOT APPROACH CAPTURED
AUTOPILOT APPROACH HOLD
AUTOPILOT APPROACH IS LOCALIZER
AUTOPILOT ATTITUDE HOLD
AUTOPILOT AVAILABLE
AUTOPILOT AVIONICS MANAGED
AUTOPILOT BACKCOURSE HOLD
AUTOPILOT BANK HOLD
AUTOPILOT BANK HOLD REF

ADF LATLONALT
ADF NAME
ADF RADIAL
ADF RADIAL MAG
ADF SIGNAL
ADF SOUND
ADF STANDBY AVAILABLE
ADF STANDBY FREQUENCY
ADF VOLUME
AI ANTISTALL STATE
AI AUTOTRIM ACTIVE
AI AUTOTRIM ACTIVE AGAINST PLAYER
AI CONTROLS
AI CURSOR MODE ACTIVE
AILERON AVERAGE DEFLECTION
AILERON LEFT DEFLECTION
AILERON LEFT DEFLECTION PCT
AILERON POSITION
AILERON RIGHT DEFLECTION
AILERON RIGHT DEFLECTION PCT
AILERON TRIM
AILERON TRIM DISABLED
AILERON TRIM PCT
AIRCRAFT WIND X
AIRCRAFT WIND Y
AIRCRAFT WIND Z
AIRSPEED BARBER POLE
AIRSPEED INDICATED
AIRSPEED MACH
AIRSPEED SELECT INDICATED OR TRUE
AIRSPEED TRUE
AIRSPEED TRUE CALIBRATE
AIRSPEED TRUE RAW
ALTERNATE STATIC SOURCE OPEN
ALTERNATOR BREAKER PULLED
ALTERNATOR CONNECTION ON
AMBIENT DENSITY
AMBIENT IN CLOUD
AMBIENT PRECIP RATE
AMBIENT PRECIP STATE
AMBIENT PRESSURE
AMBIENT TEMPERATURE
AMBIENT VISIBILITY
AMBIENT WIND DIRECTION
AMBIENT WIND VELOCITY
AMBIENT WIND X
AMBIENT WIND Y
AMBIENT WIND Z
ANEMOMETER PCT RPM
ANGLE OF ATTACK INDICATOR
ANIMATION DELTA TIME
ANNUNCIATOR SWITCH **Deprecated**

ALTITUDE
ATC FLIGHTPLAN DIFF ALT
ATC FLIGHTPLAN DIFF DISTANCE
ATC FLIGHTPLAN DIFF HEADING
ATC FLIGHT NUMBER
ATC HEAVY
ATC ID
ATC IFR FP TO REQUEST
ATC MODEL
ATC ON PARKING SPOT
ATC PREVIOUS WAYPOINT ALTITUDE
ATC RUNWAY AIRPORT NAME
ATC RUNWAY DISTANCE
ATC RUNWAY END DISTANCE
ATC RUNWAY HEADING DEGREES TRUE
ATC RUNWAY LENGTH
ATC RUNWAY RELATIVE POSITION X
ATC RUNWAY RELATIVE POSITION Y
ATC RUNWAY RELATIVE POSITION Z
ATC RUNWAY SELECTED
ATC RUNWAY START DISTANCE
ATC RUNWAY TDPOINT RELATIVE POSITION X
ATC RUNWAY TDPOINT RELATIVE POSITION Y
ATC RUNWAY TDPOINT RELATIVE POSITION Z
ATC RUNWAY WIDTH
ATC SUGGESTED MIN RWY LANDING
ATC SUGGESTED MIN RWY TAKEOFF
ATC TAXIPATH DISTANCE
ATC TYPE
ATTITUDE BARS POSITION
ATTITUDE CAGE
ATTITUDE INDICATOR BANK DEGREES
ATTITUDE INDICATOR PITCH DEGREES
AUDIO PANEL AVAILABLE
AUDIO PANEL VOLUME
AUTOBRAKES ACTIVE
AUTOPILOT AIRSPEED ACQUISITION
AUTOPILOT AIRSPEED HOLD
AUTOPILOT AIRSPEED HOLD CURRENT
AUTOPILOT AIRSPEED HOLD VAR
AUTOPILOT AIRSPEED MAX CALCULATED
AUTOPILOT AIRSPEED MIN

AUTOPILOT CRUISE SPEED HOLD
AUTOPILOT DEFAULT PITCH MODE
AUTOPILOT DEFAULT ROLL MODE
AUTOPILOT DISENGAGED
AUTOPILOT FLIGHT DIRECTOR ACTIVE
AUTOPILOT FLIGHT DIRECTOR BANK
AUTOPILOT FLIGHT DIRECTOR BANK EX1
AUTOPILOT FLIGHT DIRECTOR PITCH
AUTOPILOT FLIGHT DIRECTOR PITCH EX1
AUTOPILOT FLIGHT LEVEL CHANGE
AUTOPILOT GLIDESLOPE ACTIVE
AUTOPILOT GLIDESLOPE ARM
AUTOPILOT GLIDESLOPE HOLD
AUTOPILOT HEADING LOCK
AUTOPILOT HEADING LOCK DIR
AUTOPILOT HEADING MANUALLY TUNABLE
AUTOPILOT HEADING SLOT INDEX
AUTOPILOT MACH HOLD
AUTOPILOT MACH HOLD VAR
AUTOPILOT MANAGED INDEX **Deprecated**
AUTOPILOT MANAGED SPEED IN MACH
AUTOPILOT MANAGED THROTTLE ACTIVE
AUTOPILOT MASTER
AUTOPILOT MAX BANK
AUTOPILOT MAX BANK ID
AUTOPILOT MAX SPEED HOLD **Deprecated**
AUTOPILOT NAV1 LOCK
AUTOPILOT NAV SELECTED
AUTOPILOT PITCH HOLD
AUTOPILOT PITCH HOLD REF
AUTOPILOT RPM HOLD
AUTOPILOT RPM HOLD VAR
AUTOPILOT RPM SLOT INDEX
AUTOPILOT SPEED SETTING **Deprecated**
AUTOPILOT SPEED SLOT INDEX
AUTOPILOT TAKEOFF POWER ACTIVE
AUTOPILOT THROTTLE ARM
AUTOPILOT THROTTLE MAX THRUST
AUTOPILOT VERTICAL HOLD
AUTOPILOT VERTICAL HOLD

ANTISKID BRAKES ACTIVE
APPLY HEAT TO SYSTEMS
APU BLEED PRESSURE RECEIVED BY ENGINE
APU_GENERATOR_ACTIVE
APU_GENERATOR_SWITCH
APU_ON_FIRE_DETECTED
APU_PCT_RPM
APU_PCT_STARTER
APU_SWITCH
APU_VOLTS
CALCULATED
AUTOPILOT ALT RADIO MODE
AUTOPILOT ALTITUDE ARM
AUTOPILOT ALTITUDE LOCK VAR
AUTOPILOT ALTITUDE MANUALLY TUNABLE
AUTOPILOT ALTITUDE SLOT INDEX
AUTOPILOT APPROACH ACTIVE
AUTOPILOT APPROACH ARM
VAR
AUTOPILOT VS SLOT INDEX
AUTOPILOT WING LEVELER
AUTOPILOT YAW DAMPER
AUTOTHROTTLE ACTIVE
AUTO BRAKE SWITCH CB
AUTO COORDINATION
AUX WHEEL ROTATION ANGLE
AUX WHEEL RPM
AVIONICS MASTER SWITCH

B

BAGGAGELOADER ANGLE CURRENT
BAGGAGELOADER ANGLE TARGET
BAGGAGELOADER END RAMP Y
BAGGAGELOADER END RAMP Z
BAGGAGELOADER PIVOT Y
BAGGAGELOADER PIVOT Z
BARBER_POLE_MACH
BAROMETER_PRESSURE
BATTERY_BREAKER_PULLED
BATTERY_CONNECTION_ON
BETA_DOT
BLAST_SHIELD_POSITION
BLEED_AIR_APU
BLEED AIR ENGINE
BLEED AIR SOURCE CONTROL
BOARDINGRAMP ELEVATION CURRENT
BOARDINGRAMP ELEVATION TARGET
BOARDINGRAMP END POSITION Y
BOARDINGRAMP END POSITION Z
BOARDINGRAMP ORIENTATION CURRENT
BOARDINGRAMP ORIENTATION TARGET
BOARDINGRAMP START POSITION Y
BOARDINGRAMP START POSITION Z
BOMB_AMMO **Deprecated**
BRAKE DEPENDENT HYDRAULIC PRESSURE
BRAKE INDICATOR
BRAKE LEFT POSITION
BRAKE LEFT POSITION EX1
BRAKE PARKING INDICATOR
BRAKE PARKING POSITION
BRAKE RIGHT POSITION
BRAKE RIGHT POSITION EX1
BREAKER_ADF
BREAKER_ALTFLD
BREAKER_AUTOPILOT
BREAKER_AVNBUS1
BREAKER_AVNBUS2
BREAKER_AVNFAN
BREAKER_FLAP
BREAKER_GPS
BREAKER_INST
BREAKER_INSTLTS
BREAKER_LTS_PWR **Deprecated**
BREAKER_NAVCOM1
BREAKER_NAVCOM2
BREAKER_NAVCOM3
BREAKER_TURNCOORD
BREAKER_WARN
BREAKER_XPNDR
BUS_BREAKER_PULLED
BUS_CONNECTION_ON
BUS_LOOKUP_INDEX

C

CABIN NO SMOKING ALERT SWITCH
CABIN SEATBELTS ALERT SWITCH
CABLE_CAUGHT_BY_TAILHOOK
CAMERA_ACTION_COCKPIT_VIEW_CHASE
CAMERA_ACTION_COCKPIT_VIEW_SAVE
CAMERA_GAMEPLAY_PITCH_YAW
CAMERA_REQUEST_ACTION
CAMERA_STATE
CAMERA_SUBSTATE
CAMERA_VIEW_TYPE_AND_INDEX
CAMERA_VIEW_TYPE_AND_INDEX_MAX
CANNON_AMMO **Deprecated**
CANOPY OPEN
CARB HEAT AVAILABLE
CG_PERCENT
CG_PERCENT_LATERAL
CHASE_CAMERA_HEADLOOK
CHASE_CAMERA_MOMENTUM
CHASE_CAMERA_SPEED
CHASE_CAMERA_ZOOM
CHASE_CAMERA_ZOOM_SPEED
CIRCUIT_AUTOPILOT_ON
CIRCUIT_AUTO_BRAKES_ON
CIRCUIT AUTO FEATHER ON
CIRCUIT_AVIONICS_ON
CIRCUIT_BREAKER_PULLED
CIRCUIT_CONNECTION_ON
CIRCUIT_FLAP_MOTOR_ON
CIRCUIT_GEAR_MOTOR_ON
CIRCUIT_GEAR_WARNING_ON
COM ACTIVE BEARING
COM ACTIVE DISTANCE
COM ACTIVE FREQUENCY
COM ACTIVE FREQ IDENT
COM ACTIVE FREQ TYPE
COM ACTIVE LATLONALT **Deprecated**
COM AVAILABLE
COM LATLONALT
COM RECEIVE
COM RECEIVE ALL
COM RECEIVE EX1
COM SPACING MODE
COM STANDBY FREQUENCY
COM STANDBY FREQ IDENT
COM STANDBY FREQ TYPE

CATAPULT_STROKE_POSITION
CATEGORY
CATERINGTRUCK_AIRCRAFT_DOOR_CONTACT_OFFSET_Z
CATERINGTRUCK_ELEVATION_CURRENT
CATERINGTRUCK_ELEVATION_TARGET
CATERINGTRUCK_OPENING_CURRENT
CATERINGTRUCK_OPENING_TARGET
CENTER_WHEEL_ROTATION_ANGLE
CENTER WHEEL RPM
CG_AFT_LIMIT
CG FEET
CG FEET AFT LIMIT
CG FEET LATERAL
CG FEET LATERAL LEFT LIMIT
CG FEET LATERAL RIGHT LIMIT
CG FEET FWD LIMIT
CG_FWD_LIMIT
CG_MAX_MACH **Deprecated**
CG_MIN_MACH **Deprecated**
CIRCUIT_GENERAL_PANEL_ON
CIRCUIT_HYDRAULIC_PUMP_ON
CIRCUIT_MARKER_BEACON_ON
CIRCUIT_NAVCOM_ON1
CIRCUIT_NAVCOM_ON2
BREAKER_NAVCOM3
CIRCUIT_ON
CIRCUIT_PITOT_HEAT_ON
CIRCUIT_POWER_SETTING
CIRCUIT_PROP_SYNC_ON
CIRCUIT_STANDBY_VACUUM_ON
CIRCUIT_SWITCH_ON
COCKPIT_CAMERA_HEADLOOK
COCKPIT_CAMERA_HEIGHT
COCKPIT_CAMERA_INSTRUMENT_ACTIVE
COCKPIT_CAMERA_MOMENTUM
COCKPIT_CAMERA_SIDE
COCKPIT_CAMERA_SPEED
COCKPIT_CAMERA_UPPER_POSITION
COCKPIT_CAMERA_ZOOM
COCKPIT_CAMERA_ZOOM_SPEED
COLLECTIVE POSITION
COM1 STORED FREQUENCY
COM2 STORED FREQUENCY
COM3 STORED FREQUENCY
COM STATUS
COM TEST
COM TRANSMIT
COM VOLUME
CONCORDE_NOSE_ANGLE **Deprecated**
CONCORDE_VISOR_NOSE_HANDLE **Deprecated**
CONCORDE_VISOR_POSITION_PERCENT **Deprecated**
CONTACT_POINT_COMPRESSION
CONTACT_POINT_IS_ON_GROUND
CONTACT_POINT_IS_SKIDDING
CONTACT_POINT_POSITION
CONTACT_POINT_SKIDDING_FACTOR
CONTACT_POINT_WATER_DEPTH
CONTRAILS_CONDITIONS_MET
CONTROLLABLE
COPILOT TRANSMITTER TYPE
COPILOT TRANSMITTING
COWL FLAPS **Deprecated**
CRASH FLAG
CRASH SEQUENCE

D

DECISION_ALTITUDE_MSL
DECISION_HEIGHT
DELEGATE CONTROLS TO AI
DELTA HEADING RATE
DENSITY ALTITUDE
DESIGN_CRUISE_ALT
DESIGN_SPAWN_ALTITUDE_CRUISE
DESIGN_SPAWN_ALTITUDE_DESCENT
DESIGN_SPEED_CLIMB
DESIGN_SPEED_MIN_ROTATION
DESIGN_SPEED_VC
DESIGN_SPEED_VS0
DESIGN_SPEED_VS1
DESIGN_TAKEOFF_SPEED
DISK BANK ANGLE
DISK BANK PCT
DISK CONING PCT
DISK PITCH ANGLE
DISK PITCH PCT
DME SOUND
DRONE_CAMERA_FOCUS
DRONE_CAMERA_FOCUS_MODE
DRONE_CAMERA_FOLLOW
DRONE_CAMERA_FOV
DRONE_CAMERA_LOCKED
DRONE_CAMERA_SPEED_ROTATION
DRONE_CAMERA_SPEED_TRAVELLING
DROPPABLE OBJECTS COUNT
DROPPABLE OBJECTS TYPE
DROPPABLE OBJECTS UI NAME
DYNAMIC_PRESSURE

E

ELECTRICAL_AVIONICS_BUS_AMPS
ELECTRICAL_AVIONICS_BUS_VOLTAGE
ELECTRICAL_BATTERY_BUS_AMPS
ELECTRICAL_BATTERY_BUS_VOLTAGE
ELECTRICAL_BATTERY_ESTIMATED_CAPACITY_PCT
ELECTRICAL_BATTERY_LOAD
ELECTRICAL_BATTERY_VOLTAGE
ELECTRICAL_GENALT_BUS_AMPS
ELECTRICAL_GENALT_BUS_VOLTAGE
ELECTRICAL_GENALT_LOAD
ELECTRICAL_HOT_BATTERY_BUS_AMPS
ELECTRICAL_HOT_BATTERY_BUS_VOLTAGE
EMPTY_WEIGHT_PITCH_MOI
EMPTY_WEIGHT_ROLL_MOI
EMPTY_WEIGHT_YAW_MOI
ENGINE CONTROL SELECT
ENGINE_MIXURE_AVAILABLE **Deprecated**
ENGINE PRIMER
ENGINE TYPE
ENG ANTI ICE
ENG COMBUSTION
ENG CYLINDER HEAD TEMPERATURE
ENG OIL TEMPERATURE
ENG ON FIRE
ENG PRESSURE RATIO
ENG_PRESSURE_RATIO_GES **Deprecated**
ENG ROTOR RPM
ENG RPM ANIMATION PERCENT
ENG RPM SCALER **Deprecated**
ENG TORQUE
ENG TORQUE PERCENT
ENG_TRANSMISSION_PRESSURE **Deprecated**

ELECTRICAL_MAIN_BUS_AMPS
ELECTRICAL_MAIN_BUS_VOLTAGE
ELECTRICAL_MASTER_BATTERY
ELECTRICAL_OLD_CHARGING_AMPS **Deprecated**
ELECTRICAL_TOTAL_LOAD_AMPS
ELEVATOR DEFLECTION
ELEVATOR DEFLECTION PCT
ELEVATOR POSITION
ELEVATOR TRIM DISABLED
ELEVATOR TRIM DOWN LIMIT
ELEVATOR TRIM INDICATOR
ELEVATOR TRIM NEUTRAL
ELEVATOR TRIM PCT
ELEVATOR TRIM POSITION
ELEVATOR TRIM UP LIMIT
ELEVON DEFLECTION
ELT ACTIVATED
EMPTY_WEIGHT
EMPTY_WEIGHT_CROSS_COUPLED
ENG ELECTRICAL LOAD **Deprecated**
ENG EXHAUST GAS TEMPERATURE
ENG FAILED
ENG FUEL FLOW BUG POSITION
ENG FUEL FLOW GPH
ENG FUEL FLOW PPH
ENG FUEL FLOW PPH SSL **Deprecated**
ENG FUEL PRESSURE **Deprecated**
ENG HYDRAULIC PRESSURE
ENG HYDRAULIC QUANTITY
ENG MANIFOLD PRESSURE
ENG MAX RPM
ENG N1 RPM
ENG N2 RPM
ENG OIL PRESSURE
ENG OIL QUANTITY
ENG TRANSMISSION TEMPERATURE **Deprecated**
ENG TURBINE TEMPERATURE **Deprecated**
ENG VIBRATION
ESTIMATED_CRUISE_SPEED
ESTIMATED FUEL FLOW
EXIT OPEN **Deprecated**
EXIT POSX **Deprecated**
EXIT POSY **Deprecated**
EXIT POSZ **Deprecated**
EXIT TYPE **Deprecated**
EXTERNAL_POWER_AVAILABLE
EXTERNAL_POWER_BREAKER_PULLE
EXTERNAL_POWER_CONNECTION_O
EXTERNAL_POWER_ON
EXTERNAL SYSTEM VALUE
EYEPOINT_POSITION

F

FIRE_BOTTLE_DISCHARGED
FIRE_BOTTLE_SWITCH
FLAP_DAMAGE_BY_SPEED
FLAP_POSITION_SET
FLAP_SPEED_EXCEEDED
FLAPS_AVAILABLE
FLAPS_EFFECTIVE_HANDLE_INDEX
FLAPS_HANDLE_INDEX
FLAPS_HANDLE_PERCENT
FLAPS_NUM_HANDLE_POSITIONS
FLARM AVAILABLE
FLARM THREAT BEARING
FLARM THREAT DATA
FLARM THREAT DISTANCE
FLARM THREAT HEADING
FLARM THREAT RELATIVE ALTITUDE
FLARM THREAT TIME TO COLLISION
FLARM THREAT VERTICAL BEARING
FLY_ASSISTANT_CANCEL_DESTINATION
FLY_ASSISTANT_CANCEL_DESTINATION_SELECT
FLY_ASSISTANT_COM_AI_LOCKED
FLY_ASSISTANT_HAVE_DESTINATION
FLY_ASSISTANT_LANDING_SPEED
FLY_ASSISTANT_LANDING_SPEED_DISPLAY
FLY_ASSISTANT_NEAREST_CATEGORY
FLY_ASSISTANT_NEAREST_COUNT
FLY_ASSISTANT_NEAREST_METADATA **Deprecated**
FLY_ASSISTANT_NEAREST_NAME
FLY_ASSISTANT_NEAREST_SELECTED
FLY_ASSISTANT_RIBBONS_ACTIVE
FLY_ASSISTANT_SET_AS_DESTINATION
FLY_ASSISTANT_STALL_SPEED
FLY_BY_WIRE_ELAC_SWITCH
FLY_BY_WIRE_FAC_FAILED
FLY_BY_WIRE_FAC_SWITCH
FLY_BY_WIRE_SEC_FAILED
FLY_BY_WIRE_SEC_SWITCH
FOLDING_WING_HANDLE_POSITION
FOLDING_WING_LEFT_PERCENT
FOLDING_WING_RIGHT_PERCENT
FUELSYSTEM_ENGINE_PRESSURE
FUELSYSTEM_JUNCTION_SETTING
FUELSYSTEM_LINE_FUEL_FLOW
FUELSYSTEM_LINE_FUEL_LEVEL
FUELSYSTEM_LINE_FUEL_PRESSURE
FUELSYSTEM_PUMP_ACTIVE
FUELSYSTEM_PUMP_SWITCH
FUELSYSTEM_TANK_CAPACITY
FUELSYSTEM_TANK_LEVEL
FUELSYSTEM_TANK_QUANTITY
FUELSYSTEM_TANK_TOTAL_QUANTITY
FUELSYSTEM_TANK_WEIGHT
FUELSYSTEM_TRIGGER_STATUS
FUELSYSTEM_VALVE_OPEN
FUELSYSTEM_VALVE_SWITCH
FUELTRUCK HOSE DEPLOYED
FUELTRUCK HOSE END POSX
FUELTRUCK HOSE END POSZ
FUELTRUCK MODE HOSE END RELATIVE HEADING
FUEL CROSS FEED
FUEL DUMP ACTIVE
FUEL DUMP SWITCH
FUEL LEFT CAPACITY
FUEL LEFT QUANTITY
FUEL PUMP
FUEL RIGHT CAPACITY
FUEL TANK CENTER QUANTITY
FUEL TANK CENTER2 CAPACITY
FUEL TANK CENTER2 LEVEL
FUEL TANK CENTER2 QUANTITY
FUEL TANK CENTER3 CAPACITY
FUEL TANK CENTER3 LEVEL
FUEL TANK CENTER3 QUANTITY
FUEL TANK EXTERNAL1 CAPACITY
FUEL TANK EXTERNAL1 LEVEL
FUEL TANK EXTERNAL1 QUANTITY
FUEL TANK EXTERNAL2 CAPACITY
FUEL TANK EXTERNAL2 LEVEL
FUEL TANK EXTERNAL2 QUANTITY
FUEL TANK LEFT AUX CAPACITY
FUEL TANK LEFT AUX LEVEL
FUEL TANK LEFT AUX QUANTITY
FUEL TANK LEFT MAIN CAPACITY
FUEL TANK LEFT MAIN LEVEL
FUEL TANK LEFT MAIN QUANTITY
FUEL TANK LEFT TIP CAPACITY
FUEL TANK LEFT TIP LEVEL
FUEL TANK LEFT TIP QUANTITY
FUEL TANK RIGHT AUX CAPACITY
FUEL TANK RIGHT AUX LEVEL
FUEL TANK RIGHT AUX QUANTITY
FUEL TANK RIGHT MAIN CAPACITY
FUEL TANK RIGHT MAIN LEVEL
FUEL TANK RIGHT MAIN QUANTITY
FUEL TANK RIGHT TIP CAPACITY
FUEL TANK RIGHT TIP LEVEL
FUEL TANK RIGHT TIP QUANTITY
FUEL TANK SELECTOR
FUEL TOTAL CAPACITY
FUEL TOTAL QUANTITY
FUEL TOTAL QUANTITY WEIGHT

FLY_ASSISTANT_STALL_SPEED
FLY_ASSISTANT_STALL_SPEED_DISPLAY_MODE
FLY_ASSISTANT_TAKEOFF_SPEED
FLY_ASSISTANT_TAKEOFF_SPEED_DISPLAY_MODE
FLY_ASSISTANT_TAKEOFF_SPEED_ESTIMATED
FLY_BY_WIRE_ALPHA_PROTECTION
FLY_BY_WIRE_ELAC_FAILED
FUEL_RIGHT_CAPACITY
FUEL_RIGHT_QUANTITY
FUEL_SELECTED_QUANTITY
FUEL_SELECTED_QUANTITY_PERCENT
FUEL_SELECTED_TRANSFER_MODE
FUEL_TANK_CENTER_CAPACITY
FUEL_TANK_CENTER_LEVEL
FUEL_TOTAL_QUANTITY_WEIGHT
FUEL_TRANSFER_PUMP_ON
FUEL_WEIGHT_PER_GALLON
FULL_THROTTLE_THRUST_TO_WEIGHT_RATIO

G

GAMEPLAY_CAMERA_FOCUS
GEAR_ANIMATION_POSITION
GEAR_AUX_POSITION
GEAR_AUX_STEER_ANGLE
GEAR_AUX_STEER_ANGLE_PCT
GEAR_CENTER_POSITION
GEAR_CENTER_STEER_ANGLE
GEAR_CENTER_STEER_ANGLE_PCT
GEAR_DAMAGE_BY_SPEED
GEAR_EMERGENCY_HANDLE_POSITION
GEAR_HANDLE_POSITION
GEAR_HYDRAULIC_PRESSURE
GEAR_IS_ON_GROUND
GEAR_IS_SKIDDING
GEAR_LEFT_POSITION
GEAR_LEFT_STEER_ANGLE
GEAR_LEFT_STEER_ANGLE_PCT
GEAR_POSITION
GEAR_RIGHT_POSITION
GEAR_RIGHT_STEER_ANGLE
GEAR_RIGHT_STEER_ANGLE_PCT
GEAR SKIDDING FACTOR
GEAR_SPEED_EXCEEDED
GEAR_STEER_ANGLE
GEAR_STEER_ANGLE_PCT
GEAR_TAIL_POSITION **Deprecated**
GEAR_TOTAL_PCT_EXTENDED
GEAR_WARNING
GEAR_WATER_DEPTH
GENERAL_ENG_ANTI_ICE_POSITION
GENERAL ENG COMBUSTION
GENERAL_ENG_COMBUSTION_EX1
GENERAL_ENG_COMBUSTION_SOUND PERCENT
GENERAL_ENG_DAMAGE_PERCENT
GENERAL ENG ELAPSED TIME
GENERAL_ENG_EXHAUST_GAS_TEMPERATURE
GENERAL ENG FAILED
GENERAL ENG FIRE DETECTED
GENERAL ENG FUEL PRESSURE
GENERAL ENG FUEL PUMP ON
GENERAL_ENG_FUEL_PUMP_SWITCH
GENERAL_ENG_HOBBS_ELAPSED_TIME
GENERAL_ENG_MASTER_ALTERNATOR
GENERAL_ENG_MAX_REACHED_RPM
GENERAL_ENG_MIXTURE_LEVER_POSITION
GENERAL_ENG_OIL_LEAKED_PERCENT
GENERAL ENG OIL PRESSURE
GENERAL_ENG_OIL_TEMPERATURE
GENERAL ENG PCT MAX RPM
GENERAL_ENG_PROPELLER_LEVER POSITION
GENERAL_ENG_REVERSE_THRUST ENGAGED
GENERAL ENG RPM
GENERAL ENG STARTER
GENERAL ENG STARTER ACTIVE
GENERAL_ENG_THROTTLE_LEVER POSITION
GENERAL_ENG_THROTTLE_MANAGED MODE
GLASSCOCKPIT_AUTOMATIC_BRIGHTNESS
GPS_APPROACH_AIRPORT_ID
GPS_APPROACH_APPROACH_ID
GPS_APPROACH_APPROACH_INDEX
GPS_APPROACH_APPROACH_TYPE
GPS_APPROACH_IS_FINAL
GPS_APPROACH_IS_MISSED
GPS_APPROACH_IS_WP_RUNWAY
GPS_APPROACH_MODE
GPS_APPROACH_SEGMENT_TYPE
GPS_APPROACH_TIMEZONE_DEVIATION
GPS_APPROACH_TRANSITION_ID
GPS_APPROACH_TRANSITION_INDEX
GPS_APPROACH_WP_COUNT
GPS_APPROACH_WP_INDEX
GPS_APPROACH_WP_TYPE
GPS CDI SCALING
GPS_COURSE_TO_STEER
GPS_DRIVES_NAV1
GPS_ETA
GPS_ETE
GPS HAS GLIDEPATH
GPS_IS_ACTIVE_FLIGHT_PLAN
GPS_IS_ACTIVE_WAY_POINT
GPS_IS_ACTIVE_WP_LOCKED
GPS_IS_APPROACH_ACTIVE
GPS_IS_APPROACH_LOADED
GPS_IS_ARRIVED
GPS_IS_DIRECTTO_FLIGHTPLAN
GPS_MAGVAR
GPS_OBS_ACTIVE
GPS_OBS_VALUE
GPS_OVERRIDDEN
GPS_POSITION_ALT
GPS_POSITION_LAT
GPS_POSITION_LON
GPS_TARGET_ALTITUDE
GPS_TARGET_DISTANCE
GPS_VERTICAL_ANGLE
GPS_VERTICAL_ANGLE_ERROR
GPS_VERTICAL_ERROR
GPS_WP_BEARING
GPS_WP_CROSS_TRK
GPS_WP_DESIRED_TRACK
GPS_WP_DISTANCE
GPS_WP_ETA
GPS_WP_ETE
GPS_WP_NEXT_ALT
GPS_WP_NEXT_ID
GPS_WP_NEXT_LAT
GPS_WP_NEXT_LON
GPS_WP_PREV_ALT
GPS_WP_PREV_ID
GPS_WP_PREV_LAT
GPS_WP_PREV_LON
GPS_WP_PREV_VALID
GPS_WP_TRACK_ANGLE_ERROR
GPS_WP_TRUE_BEARING
GPS_WP_TRUE_REQ_HDG
GPS_WP_VERTICAL_SPEED
GPWS_SYSTEM_ACTIVE
GPWS_WARNING
GROUNDPOWERUNIT_HOSE_DEPLOYED
GROUNDPOWERUNIT_HOSE_END POSX
GROUNDPOWERUNIT_HOSE_END POSZ

GENERAL ENG FUEL PUMP SWITCH EX1
GENERAL ENG FUEL USED SINCE START
GENERAL ENG FUEL VALVE
GENERAL ENG GENERATOR ACTIVE
GENERAL ENG GENERATOR
GPS FLIGHTPLAN TOTAL DISTANCE
GPS FLIGHT PLAN WP COUNT
GPS FLIGHT PLAN WP INDEX
GPS GROUND MAGNETIC TRACK
GPS GROUND SPEED
GPS GROUND TRUE HEADING
GPS GROUND TRUE TRACK
GPS GSI SCALING
GROUNDPOWERUNIT HOSE
END RELATIVE HEADING
GROUND ALTITUDE
GROUND VELOCITY
GUN_AMMO ***Deprecated***
GYRO DRIFT ERROR
G FORCE
G LIMITER SETTING

H

HAS STALL PROTECTION
HAND ANIM STATE
HEADING INDICATOR
HELICOPTER MASTER THROTTLE POSITION
HOLDBACK BAR INSTALLED
HSI BEARING
HSI BEARING VALID
HSI CDI NEEDLE
HSI CDI NEEDLE VALID
HSI DISTANCE
HSI GSI NEEDLE
HSI GSI NEEDLE VALID
HSI HAS LOCALIZER
HSI SPEED
HSI STATION IDENT
HSI TF FLAGS
HYDRAULIC PRESSURE
HYDRAULIC RESERVOIR PERCENT
HYDRAULIC SWITCH
HYDRAULIC SYSTEM INTEGRITY

I

IDLE ANIMATION ID
INCIDENCE ALPHA
INCIDENCE BETA
INDICATED ALTITUDE
INDICATED ALTITUDE CALIBRATED
INDICATED ALTITUDE EX1
INDUCTOR COMPASS HEADING REF
INDUCTOR COMPASS PERCENT DEVIATION
INNER MARKER
INNER MARKER LATLONALT
INSTRUMENTS_AVAILABLE ***Deprecated***
INTERACTIVE POINT BANK
INTERACTIVE POINT HEADING
INTERACTIVE POINT JETWAY LEFT BEND
INTERACTIVE POINT JETWAY LEFT DEPLOYMENT
INTERACTIVE POINT JETWAY RIGHT BEND
INTERACTIVE POINT JETWAY RIGHT DEPLOYMENT
INTERACTIVE POINT JETWAY TOP HORIZONTAL
INTERACTIVE POINT JETWAY TOP VERTICAL
INTERACTIVE POINT GOAL
INTERACTIVE POINT OPEN
INTERACTIVE POINT PITCH
INTERACTIVE POINT POSX
INTERACTIVE POINT POSY
INTERACTIVE POINT POSZ
INTERACTIVE POINT TYPE
INTERCOM MODE
INTERCOM SYSTEM ACTIVE
IS ALTITUDE FREEZE ON
IS ANY INTERIOR LIGHT ON
IS ATTACHED TO SLING
IS ATTITUDE FREEZE ON
IS CAMERA RAY INTERSECT WITH
IS GEAR FLOATS
IS GEAR RETRACTABLE
IS GEAR SKIDS
IS GEAR SKIS
IS GEAR WHEELS
IS LATITUDE LONGITUDE FREEZE ON
IS SLEW ACTIVE
IS SLEW ALLOWED
IS TAIL DRAGGER
IS USER SIM

J

JETWAY HOOD LEFT BEND
JETWAY HOOD LEFT DEPLOYMENT
JETWAY HOOD RIGHT BEND
JETWAY HOOD RIGHT DEPLOYMENT
JETWAY HOOD TOP HORIZONTAL
JETWAY HOOD TOP VERTICAL
JETWAY MOVING
JETWAY WHEEL ORIENTATION CURRENT
JETWAY WHEEL ORIENTATION TARGET
JETWAY WHEEL SPEED

K

KOHLSMAN SETTING HG
KOHLSMAN SETTING MB
KOHLSMAN SETTING STD

L

LANDING_LIGHT_PBH
LAUNCHBAR_HELD_EXTENDED
LAUNCHBAR_POSITION
LAUNCHBAR_SWITCH
LEADING_EDGE_FLAPS_LEFT_ANGLE
LEADING_EDGE_FLAPS_LEFT_INDEX
LEADING_EDGE_FLAPS_LEFT_PERCENT
LEADING_EDGE_FLAPS_RIGHT_ANGLE
LEADING_EDGE_FLAPS_RIGHT_INDEX
LEADING_EDGE_FLAPS_RIGHT_PERCENT
LEFT_WHEEL_ROTATION_ANGLE
LEFT_WHEEL_RPM
LIGHT_BACKLIGHT_INTENSITY
LIGHT_BEACON
LIGHT_BEACON_ON
LIGHT_BRAKE_ON
LIGHT_CABIN
LIGHT_CABIN_ON
LIGHT_CABIN_POWER_SETTING
LIGHT_GLARESHIELD
LIGHT_GLARESHIELD_ON
LIGHT_GLARESHIELD_POWER_SETTING
LIGHT_GLARESHIELD_POWER_SETTING
LIGHT_HEADLIGHT_INTENSITY
LIGHT_HEAD_ON
LIGHT_LANDING
LIGHT_LANDING_ON
LIGHT_LOGO
LIGHT_LOGO_ON
LIGHT_NAV
LIGHT_NAV_ON
LIGHT_ON_STATES
LIGHT_PANEL
LIGHT_PANEL_ON
LIGHT_PANEL_POWER_SETTING
LIGHT_PEDESTRAL
LIGHT_PEDESTRAL_ON
LIGHT_PEDESTRAL_POWER_SETTING
LIGHT_POTENTIOMETER
LIGHT_RECOGNITION
LIGHT_RECOGNITION_ON
LIGHT_STATES
LIGHT_STROBE
LIGHT_STROBE_ON
LIGHT_TAXI
LIGHT_TAXI_ON
LIGHT_WING
LIGHT_WING_ON
LINEAR_CL_ALPHA

M

MACH_MAX_OPERATE
MAGNETIC_COMPASS
MAGVAR
MANUAL_FUEL_PUMP_HANDLE
MANUAL_INSTRUMENT_LIGHTS
MARKER_AVAILABLE
MARKER_BEACON_SENSITIVITY_HIGH
MARKER_BEACON_STATE
MARKER_BEACON_TEST_MUTE
MARKER_SOUND
MARSHALLER AIRCRAFT DIRECTION PARKINGSPACE
MARSHALLER AIRCRAFT DISTANCE
MARSHALLER AIRCRAFT DISTANCE DIRECTION X PARKINGSPACE
MARSHALLER AIRCRAFT DISTANCE DIRECTION Z PARKINGSPACE
MARSHALLER AIRCRAFT ENGINE SHUTDOWN
MARSHALLER AIRCRAFT HEADING PARKINGSPACE
MARSHALLER AIRCRAFT PROJECTION POINT PARKINGSPACE
MARSHALLER AIRCRAFT VELOCITY
MASTER_IGNITION_SWITCH
MAX EGT
MAX_GROSS_WEIGHT
MAX_G_FORCE
MAX OIL TEMPERATURE
MAX RATED ENGINE RPM
MIDDLE_MARKER
MIDDLE_MARKER_LATLONALT
MIN_DRAG_VELOCITY
MIN_G_FORCE
MISSION_SCORE

N

NAV_ACTIVE_FREQUENCY
NAV_AVAILABLE
NAV_BACK_COURSE_FLAGS
NAV_CDI
NAV_CLOSE_DME
NAV_CLOSE_FREQUENCY
NAV_CLOSE_IDENT
NAV_CLOSE_LOCALIZER
NAV_CLOSE_NAME
NAV_CODES
NAV_DME
NAV_DMESPEED
NAV_DME_LATLONALT
NAV_FREQUENCY
NAV_GS_LATLONALT
NAV_GS_LLAF64
NAV_HAS_CLOSE_DM
NAV_HAS_CLOSE_LOCALIZER
NAV_HAS_DME
NAV_HAS_GLIDE_SLOPE
NAV_HAS_LOCALIZER
NAV_HAS_NAV
NAV_HAS_TACAN
NAV_IDENT
NAV_LOCALIZER
NAV_LOC_AIRPORT_IDENT
NAV_LOC_RUNWAY_DESIGNATOR
NAV_LOC_RUNWAY_NUMBER
NAV_RAW_GLIDE_SLOPE
NAV_RELATIVE_BEARING_TO_STATIC
NAV_SIGNAL
NAV_SOUND
NAV_STANDBY_FREQUENCY
NAV_TOFROM
NAV_VOLUME
NAV_VOR_DISTANCE
NAV_VOR_LATLONALT
NAV_VOR_LLAF64
NEW_ELECTRICAL_SYSTEM
NEW_FUEL_SYSTEM
NOSEWHEEL_LOCK_ON
NOSEWHEEL_MAX_STEERING_ANGLE

NAV_GLIDE_SLOPE
NAV_GLIDE_SLOPE_ERROR
NAV_GLIDE_SLOPE_LENGTH
NAV_GSI
NAV_GS_FLAG
NAV_MAGVAR
NAV_NAME
NAV_OBS
NAV_RADIAL
NAV_RADIAL_ERROR
NUMBER_OF_CATAPULTS
NUMBER_OF_ENGINES
NUM_FUEL_SELECTORS
NUM SLING CABLES

O

OIL_AMOUNT **Deprecated**
ON_ANY_RUNWAY
ORNITHOPTER CONTROL LEVERS X
ORNITHOPTER CONTROL LEVERS Y
ORNITHOPTER DIVE MODE ENABLED
ORNITHOPTER GLIDE MODE ENABLED
ORNITHOPTER WINGS BRAKE ENABLED
ORNITHOPTER WINGS BRAKE ACTIVE
ORNITHOPTER WING HORIZONTAL TILT
ORNITHOPTER WING PITCH
ORNITHOPTER WING VERTICAL TILT
ORNITHOPTER WINGS BLUR
ORNITHOPTER WINGS CLUTCH STATE
ORNITHOPTER WINGS FOLDING OPERATION ID
ORNITHOPTER WORLD LIFT
OUTER_MARKER
OUTER_MARKER_LATLONALT
OVERSPEED_WARNING

P

PANEL_ANTI_ICE_SWITCH
PANEL AUTO FEATHER SWITCH
PARACHUTE OPEN
PARTIAL_PANEL_ADF
PARTIAL_PANEL_AIRSPEED
PARTIAL_PANEL_ALTIMETER
PARTIAL_PANEL_ATTITUDE
PARTIAL_PANEL_AVIONICS
PARTIAL_PANEL_COMM
PARTIAL_PANEL_COMPASS
PARTIAL_PANEL_ELECTRICAL
PARTIAL_PANEL_ENGINE
PARTIAL_PANEL_FUEL_INDICATOR
PARTIAL_PANEL_HEADING
PARTIAL_PANEL_NAV
PARTIAL_PANEL_PITOT
PARTIAL_PANEL_TRANSPONDER
PARTIAL_PANEL_TURN_COORDINATOR
PARTIAL_PANEL_VACUUM
PARTIAL_PANEL_VERTICAL_VELOCITY
PAYLOAD_STATION_COUNT
PAYLOAD_STATION_NAME
PAYLOAD_STATION_NUM_SIMOBJECTS
PAYLOAD_STATION_OBJECT
PAYLOAD_STATION_WEIGHT
PILOT_TRANSMITTER_TYPE
PILOT_TRANSMITTING
PITOT_HEAT
PITOT_HEAT_SWITCH
PITOT_ICE_PCT
PLANE_ALTITUDE
PLANE_ALT_ABOVE_GROUND
PLANE_ALT_ABOVE_GROUND_MINUS_CG
PLANE_BANK_DEGREES
PLANE_HEADING_DEGREES_GYRO
PLANE_HEADING_DEGREES_MAGNETIC
PLANE_HEADING_DEGREES_TRUE
PLANE_IN_PARKING_STATE
PLANE_LATITUDE
PLANE_LONGITUDE
PLANE_PITCH_DEGREES
PLANE_TOUCHDOWN_BANK_DEGREES
PLANE_TOUCHDOWN_HEADING_DEGREES_MAGNETIC
PLANE_TOUCHDOWN_HEADING_DEGREES_TRUE
PLANE_TOUCHDOWN_LATITUDE
PLANE_TOUCHDOWN_LONGITUDE
PLANE_TOUCHDOWN_NORMAL_VELOCITY
PLANE_TOUCHDOWN_PITCH_DEGREES
PRESSURE_ALTITUDE
PRESSURIZATION_CABIN_ALTITUDE
PRESSURIZATION_CABIN_ALTITUDE_GOAL
PRESSURIZATION_CABIN_ALTITUDE_RATE
PRESSURIZATION_DUMP_SWITCH
PRESSURIZATION_PRESSURE_DIFFERENTIAL
PROP_AUTO_CRUISE_ACTIVE
PROP_AUTO_FEATHER_ARMED
PROP_BETA
PROP BETA FORCED ACTIVE
PROP BETA FORCED POSITION
PROP_BETA_MAX
PROP_BETA_MIN
PROP_BETA_MIN_REVERSE
PROP_DEICE_SWITCH
PROP_FEATHERED
PROP_FEATHERING_INHIBIT
PROP_FEATHER_SWITCH
PROP_MAX_RPM_PERCENT
PROP_ROTATION_ANGLE
PROP_RPM
PROP_SYNC_ACTIVE
PROP_SYNC_DELTA_LEVER
PROP_THRUST
PROPELLER_ADVANCED_SELECTION
PUSHBACK ANGLE
PUSHBACK ATTACHED
PUSHBACK AVAILABLE
PUSHBACK CONTACTX
PUSHBACK CONTACTY
PUSHBACK CONTACTZ
PUSHBACK STATE
PUSHBACK WAIT

R

RADIOS_AVAILABLE
RECIP_ENG_NITROUS_TANK_MAX_QUANTITY
ROTOR BRAKE ACTIVE

RADIO_HEIGHT
RAD_INS_SWITCH
REALISM
REALISM_CRASH_DETECTION
REALISM_CRASH_WITH_OTHERS
RECIP_CARBURETOR_TEMPERATURE
RECIP_CARBURETOR_TEMPERATURE
RECIP_ENG_ALTERNATE_AIR_POSITION
RECIP_ENG_ANTIDETONATION_FLOW_RATE
RECIP_ENG_ANTIDETONATION_TANK_MAX_QUANTITY
RECIP_ENG_ANTIDETONATION_TANK_QUANTITY
RECIP_ENG_ANTIDETONATION_TANK_VALVE
RECIP_ENG_BRAKE_POWER
RECIP_ENG_COOLANT_RESERVOIR_PERCENT
RECIP_ENG_COWL_FLAP_POSITION
RECIP_ENG_CYLINDER_HEAD_TEMPERATURE
RECIP_ENG_CYLINDER_HEALTH
RECIP_ENG_DETONATING
RECIP_ENG_EMERGENCY_BOOST_ACTIVE
RECIP_ENG_EMERGENCY_BOOST_ELAPSED_TIME
RECIP_ENG_ENGINE_MASTER_SWITCH
RECIP_ENG_FUEL_AVAILABLE
RECIP_ENG_FUEL_FLOW
RECIP_ENG_FUEL_NUMBER_TANKS_USED
RECIP_ENG_FUEL_TANKS_USED
RECIP_ENG_FUEL_TANK_SELECTOR
RECIP_ENG_GLOW_PLUG_ACTIVE
RECIP_ENG_LEFT_MAGNETO
RECIP_ENG_MANIFOLD_PRESSURE
RECIP_ENG_NITROUS_TANK_QUANTITY
RECIP_ENG_NITROUS_TANK_VALVE
RECIP_ENG_NUM_CYLINDERS
RECIP_ENG_NUM_CYLINDERS_FAILED
RECIP_ENG_PRIMER
RECIP_ENG_RADIATOR_TEMPERATURE
RECIP_ENG_RIGHT_MAGNETO
RECIP_ENG_STARTER_TORQUE
RECIP_ENG_TURBINE_INLET_TEMPERATURE
RECIP_ENG_TURBOCHARGER_FAILED
RECIP_ENG_WASTEGATE_POSITION
RECIP_VALVETURE_RATIO
REJECTED_TAKEOFF_BRAKES_ACTIVE
RELATIVE_WIND_VELOCITY_BODY_X
RELATIVE_WIND_VELOCITY_BODY_Y
ROTATION_ACCELERATION_BODY_Z

RETRACT_FLOAT_SWITCH
RETRACT_LEFT_FLOAT_EXTENDED
RETRACT_RIGHT_FLOAT_EXTENDED
RIGHT_WHEEL_ROTATION_ANGLE
RIGHT_WHEEL_RPM
ROCKET_AMMO **Deprecated**
ROTATION_ACCELERATION_BODY_X
ROTATION_ACCELERATION_BODY_Y
ROTATION_ACCELERATION_BODY_Z
ROTATION_VELOCITY_BODY_X
ROTATION_VELOCITY_BODY_Y
ROTATION_VELOCITY_BODY_Z
ROTOR_BRAKE_HANDLE_POS
ROTOR_CHIP_DETECTED **Deprecated**
ROTOR_CLUTCH_ACTIVE
ROTOR_CLUTCH_SWITCH_POS
ROTOR_COLLECTIVE_BLADE_PITCH_PCT
ROTOR_CYCLIC_BLADE_MAX_PITCH_POSITION
ROTOR_CYCLIC_BLADE_PITCH_PCT
ROTOR_GOV_ACTIVE
ROTOR_GOV_SWITCH_POS
ROTOR_LATERAL_TRIM_PCT
ROTOR_LONGITUDINAL_TRIM_PCT
ROTOR_ROTATION_ANGLE
ROTOR_RPM
ROTOR_RPM_PCT
ROTOR_TEMPERATURE **Deprecated**
RUDDER_DEFLECTION
RUDDER_DEFLECTION_PCT
RUDDER_PEDAL_INDICATOR
RUDDER_PEDAL_POSITION
RUDDER_POSITION
RUDDER_TRIM
RUDDER_TRIM_DISABLED
RUDDER_TRIM_PCT

S

SEA_LEVEL_PRESSURE
SELECTED_DME
SEMIBODY_LOADFACTOR_X **Deprecated**
SEMIBODY_LOADFACTOR_Y
SEMIBODY_LOADFACTOR_YDOT
SEMIBODY_LOADFACTOR_Z **Deprecated**
SHUTOFF_VALVE_PULLED
SIGMA_SQRT
SIMULATED_RADIUS
SIM_DISABLED
SIM_ON_GROUND
SIM_SHOULD_SET_ON_GROUND
SLING_ACTIVE_PAYLOAD_STATION
SLING_CABLE_BROKEN
SLING_CABLE_EXTENDED_LENGTH
SLING_HOIST_PERCENT_DEPLOYED
SLING_HOIST_SWITCH
SLING_HOOK_IN_PICKUP_MODE
SLING_OBJECT_ATTACHED
SLOPE_TO_ATC_RUNWAY
SMART_CAMERA_ACTIVE
SMART_CAMERA_INFO
SMART_CAMERA_LIST
SMART_CAMERA_LIST_DESCRIPTION
SMOKESYSTEM_AVAILABLE
SMOKE_ENABLE
SPEAKER_ACTIVE
SPOILERS_ARMED
SPOILER_AVAILABLE
SPOILERS_HANDLE_POSITION
SPOILERS_LEFT_POSITION
SPOILERS_RIGHT_POSITION
STALL_ALPHA
STALL_HORN_AVAILABLE
STALL_PROTECTION_OFF_LIMIT
STALL_PROTECTION_ON_GOAL
STALL_PROTECTION_ON_LIMIT
STALL_WARNING
STANDARD_ATM_TEMPERATURE
STATIC_CG_TO_GROUND
STATIC_PITCH
STEER_INPUT_CONTROL
STROBES_AVAILABLE
STROBE_FLASH **Deprecated**
STRUC_AIRSPEED_HOLD_PID_CONSTS
STRUC_HEADING_HOLD_PID_CONSTS
STRUCT_AMBIENT_WIND
STRUCT_BODY_ROTATION_ACCELERA
STRUCT_BODY_ROTATION_VELOCITY
STRUCT_BODY_VELOCITY
STRUCT_DAMAGEVISIBLE
STRUCT_ENGINE_POSITION
STRUCT_EYEPOINT_DYNAMIC_ANGLE
STRUCT_EYEPOINT_DYNAMIC_OFFSE
STRUCT_LATLONALT
STRUCT_LATLONALTPBH
STRUCT_PBH32
STRUCT_REALISM_VARS
STRUCT_ROTOR_POSITION
STRUCT_SURFACE_RELATIVE_VELOC
STRUCT_WORLDVELOCITY
STRUCT_WORLD_ACCELERATION
STRUCT_WORLD_ROTATION_VELOCI
STRUCTURAL_DEICE_SWITCH
STRUCTURAL_ICE_PCT
SUCTION_PRESSURE
SURFACE_CONDITION
SURFACE_INFO_VALID
SURFACE_RELATIVE_GROUND_SPEED
SURFACE_TYPE
SYSTEMS_AVAILABLE **Deprecated**

SMART_CAMERA_INFO
STRUCT_AMBIENT_WIND

T

TACAN_ACTIVE_CHANNEL
TACAN_ACTIVE_MODE
TACAN_AVAILABLE
TACAN DRIVES NAV
TACAN OBS
TACAN_STANDBY_CHANNEL
TACAN_STANDBY_MODE
TACAN STATION CDI
TACAN_STATION_DISTANCE
TACAN STATION IDENT
TACAN_STATION_LATLONALT
TACAN STATION RADIAL
TACAN STATION RADIAL ERROR
TACAN STATION TOFROM
TACAN VOLUME
TAIL ROTOR BLADE PITCH PCT
TAIL ROTOR PEDAL POSITION
TAILHOOK_HANDLE
TAILHOOK_POSITION
TAILWHEEL_LOCK_ON
THROTTLE_LOWER_LIMIT
TITLE
TOE_BRAKES_AVAILABLE
TOTAL_AIR_TEMPERATURE
TOTAL_VELOCITY
TOTAL_WEIGHT
TOTAL_WEIGHT_CROSS_COUPLED_TOR
TOTAL_WEIGHT_PITCH_MOI
TOTAL_WEIGHT_ROLL_MOI
TOTAL_WEIGHT_YAW_MOI
TOTAL_WORLD_VELOCITY
TOW_CONNECTION
TOW_RELEASE_HANDLE
TRACK_IR_ENABLE
TRAILING_EDGE_FLAPS_LEFT_ANGLE
TRAILING_EDGE_FLAPS_LEFT_INDEX
TRAILING_EDGE_FLAPS_LEFT_PERCENT
TRAILING_EDGE_FLAPS_RIGHT_ANGLE
TRAILING_EDGE_FLAPS_RIGHT_INDEX
TRAILING_EDGE_FLAPS_RIGHT_PERCENT
TRANSPONDER_AVAILABLE
TRANSPONDER_CODE
TRANSPONDER_IDENT
TRANSPONDER_STATE
TRUE_AIRSPEED_SELECTED
TURB_ENG_AFTERBURNER
TURB_ENG_AFTERBURNER_PCT_ACTIVE
TURB_ENG_AFTERBURNER_STAGE_ACTIVE
TURB_ENG_BLEED_AIR
TURB_ENG_COMMANDED_N1
TURB_ENG_CONDITION_LEVER_POSITION
TURB_ENG_CORRECTED_FF
TURB_ENG_CORRECTED_N1
TURB_ENG_CORRECTED_N2
TURB_ENG_FREE_TURBINE_TORQUE
TURB_ENG_FUEL_AVAILABLE
TURB_ENG_FUEL_EFFICIENCY_LOSS
TURB_ENG_FUEL_FLOW_PPH
TURB_ENG_HIGH_IDLE
TURB_ENG_IGNITION_SWITCH
TURB_ENG_IGNITION_SWITCH_EX1
TURB_ENG_IS_IGNITING
TURB_ENG_ITT
TURB_ENG_ITT_COOLING_EFFICIENCY_LOSS
TURB_ENG_JET_THRUST
TURB_ENG_LOW_IDLE
TURB_ENG_MASTER_STARTER_SWIT
TURB_ENG_MAX_TORQUE_PERCENT
TURB_ENG_N1
TURB_ENG_N1_LOSS
TURB_ENG_N2
TURB_ENG_NUM_TANKS_USED
TURB_ENG_PRESSURE_RATIO
TURB_ENG_PRIMARY_NOZZLE_PERC
TURB_ENG_REVERSE_NOZZLE_PERC
TURB_ENG_TANKS_USED
TURB_ENG_TANK_SELECTOR
TURB_ENG_THROTTLE_COMMANDED
TURB_ENG_THRUST_EFFICIENCY_LOSS
TURB_ENG_VIBRATION
TURN_COORDINATOR_BALL
TURN_COORDINATOR_BALL_INV
TURN_INDICATOR_RATE
TURN_INDICATOR_SWITCH
TYPICAL_DESCENT_RATE

U

UNLIMITED_FUEL
UNUSABLE_FUEL_TOTAL_QUANTITY
USER_INPUT_ENABLED

V

VARIOMETER_MAC_CREADY_SETTING
VARIOMETER NETTO
VARIOMETER RATE
VARIOMETER SPEED TO FLY
VARIOMETER_SPEED_TO_FLY
GLIDE RATIO
VARIOMETER SWITCH
VARIOMETER TOTAL ENERGY
VELOCITY_BODY_X
VELOCITY_BODY_Y
VELOCITY_BODY_Z
VELOCITY_WORLD_X
VELOCITY_WORLD_Y
VELOCITY_WORLD_Z
VERTICAL_SPEED
VISUAL MODEL RADIUS

W

WAGON_BACK_LINK_ORIENTATION
WAGON_BACK_LINK_START_POSZ
WAGON_FRONT_LINK_LENGTH
WAGON_FRONT_LINK_ORIENTATION
WAGON_FRONT_LINK_START_POSZ
WARNING_FUEL
WARNING_FUEL_LEFT
WARNING_FUEL_RIGHT
WARNING_LOW_HEIGHT
WARNING_OIL_PRESSURE
WARNING_VACUUM
WARNING_VACUUM_LEFT
WARNING_VACUUM_RIGHT
WARNING_VOLTAGE
WATER_BALLAST_EVERY_VALVE_OPEN
WATER_BALLAST_TANK_CAPACITY
WATER_BALLAST_TANK_NUMBER
WATER_BALLAST_TANK_QUANTITY
WATER_BALLAST_VALVE
WATER_BALLAST_VALVE_FLOW_RATE
WATER_LEFT_RUDDER_EXTENDED
WATER_LEFT_RUDDER_STEER_ANGL
WATER_LEFT_RUDDER_STEER_ANGL
WATER_RIGHT_RUDDER_EXTENDED
WATER_RIGHT_RUDDER_STEER_ANG
WATER_RIGHT_RUDDER_STEER_ANG
WATER_RUDDER_HANDLE_POSITION
WHEEL_ROTATION_ANGLE
WHEEL_RPM
WINDSHIELD_DEICE_SWITCH
WINDSHIELD_RAIN_EFFECT_AVAILAE
WINDSHIELD_WIND_VELOCITY
WING_AREA
WING_FLEX_PCT
WING_SPAN
WISKEY_COMPASS_INDICATION_DE

Y

YAW_STRING_ANGLE
YAW_STRING_PCT_EXTENDED
YOKE_X_INIDICATOR
YOKE_X_POSITION
YOKE_X_POSITION_WITH_AP
YOKE_Y_INIDICATOR
YOKE_Y_POSITION
YOKE_Y_POSITION_WITH_AP

Z

ZERO_LIFT_ALPHA

Note on SimVar Data Types

When accessing SimVars using the SimConnect API, you would first create a data definition using the function `SimConnect_AddToDataDefinition` and in that you would tell the simulation the data type for the variable. By default this is a 64 bit float, but for most simulation variables this can be almost any appropriate data type (that fits the unit of the SimVar) and the same data type should be used when getting or setting the value later. When a specific data type is required - for example `SIMCONNECT_DATA_LATLONALT` - this will be mentioned in the description for the variable.

***NOTE:** SimConnect has **no** BOOL data type. For this kind of value, you*

*NOTE: SimConnect has **no** BOOL data type. For this kind of value, you can use an int or a float, and the simulation will interpret 0 as `false`, and anything over 0 will be interpreted as `true`.*

Note On SimVars In Multiplayer

Previous iterations of Flight Simulator permitted the communication of SimVars between aircraft when multiplayer was active, either within a **shared cockpit** or to **all aircraft**. However Microsoft Flight Simulator 2020 does not have the same concept of shared cockpit as previous versions, as such only a small number of variables are communicated between **all aircraft**. The table below lists all of these variables, and they are also flagged within the SimVar descriptions on their respective pages. Note that the SimVars are split into two categories where:

- **Near**: these SimVars are transmitted to all aircraft within a 500m radius.
- **Far**: these SimVars are transmitted to all aircraft up to a maximum radius of 10,000m.

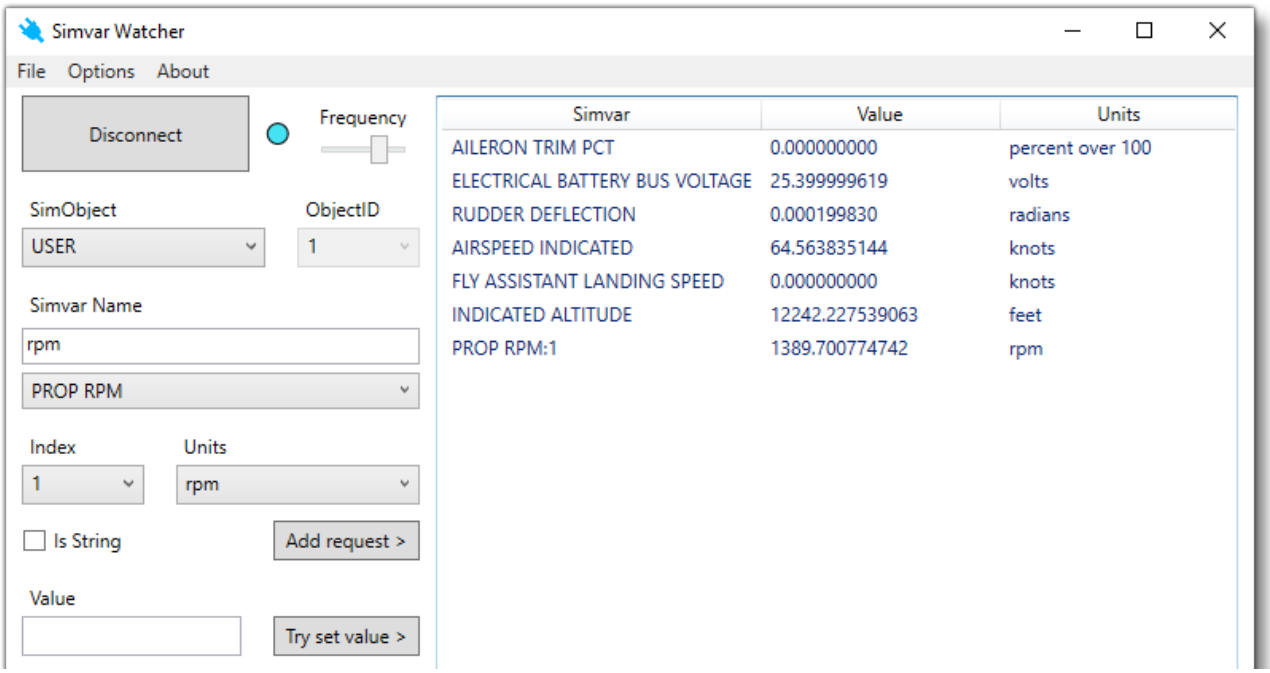
NEAR (up to 500m)	FAR (up to 10,000m)
AILERON LEFT DEFLECTION PCT	DISK BANK ANGLE
AILERON RIGHT DEFLECTION PCT	DISK BANK PCT
AILERON TRIM PCT	DISK CONING PCT
CENTER WHEEL ROTATION ANGLE	DISK PITCH ANGLE
ELEVATOR DEFLECTION PCT	DISK PITCH PCT
ENG RPM ANIMATION PERCENT	ENG ROTOR RPM
FLAPS_HANDLE_PERCENT	GENERAL ENG COMBUSTION
	GENERAL ENG COMBUSTION SOUND

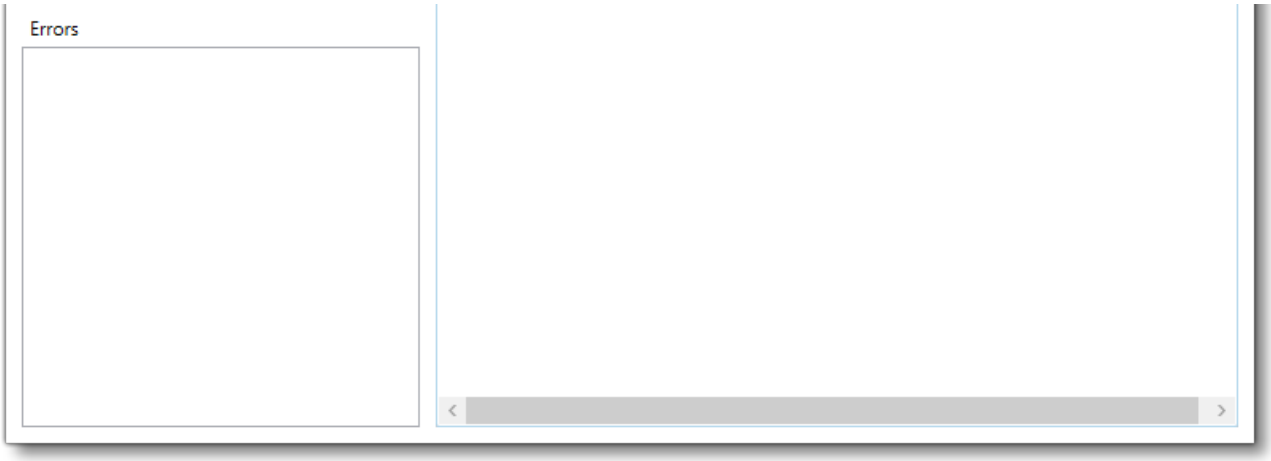
GEAR_ANIMATION_POSITION	GENERAL ENG COMBUSTION SOUND PERCENT
GEAR_CENTER_STEER_ANGLE	GENERAL ENG PCT MAX RPM
GEAR_LEFT_STEER_ANGLE	GENERAL ENG RPM
GEAR_POSITION	GROUND_VELOCITY
GEAR_RIGHT_STEER_ANGLE	LIGHT_BEACON
GEAR_STEER_ANGLE	LIGHT_LANDING
GENERAL ENG STARTER ACTIVE	LIGHT_LOGO
LEADING_EDGE_FLAPS_LEFT_PERCENT	LIGHT_NAV
LEADING_EDGE_FLAPS_RIGHT_PERCENT	LIGHT_PEDESTRAL
PROP_MAX_RPM_PERCENT	LIGHT_RECOGNITION
PROP_ROTATION_ANGLE	LIGHT_STATES
PROP_THRUST	LIGHT_STROBE
RUDDER_DEFLECTION_PCT	LIGHT_STROBE_ON
SIM ON GROUND	LIGHT_TAXI
SPOILERS_LEFT_POSITION	LIGHT_WING
SPOILERS_RIGHT_POSITION	PROP_BETA
TRAILING_EDGE_FLAPS_LEFT_PERCENT	PROP_RPM
TRAILING_EDGE_FLAPS_RIGHT_PERCENT	ROTOR COLLECTIVE BLADE PITCH PCT
TURB_ENG_AFTERBURNER	ROTOR CYCLIC BLADE PITCH PCT
TURB_ENG_AFTERBURNER_PCT_ACTIVE	ROTOR CYCLIC BLADE MAX PITCH POSITION
TURB_ENG_AFTERBURNER_STAGE_ACTIVE	ROTOR LATERAL TRIM PCT
	ROTOR LONGITUDINAL TRIM PCT

	ROTOR ROTATION ANGLE
	ROTOR RPM
	SIM ON GROUND
	TURB_ENG_N1
	TURB_ENG_N2
	TURB_ENG_REVERSE_NOZZLE_PERCENT
	TURN_INDICATOR_RATE
	VELOCITY_WORLD_X
	VELOCITY_WORLD_Y
	VELOCITY_WORLD_Z

Simvar Watcher

An essential tool when working with the various simulation variables and setting up an aircraft is the **Simvar Watcher** tool:





This tool permits you to connect to the simulation in real-time and view the values of almost any SimVar available, as well as set the SimVar to a different value (if the SimVar is settable). Find out more information on this useful tool here:

- SimvarWatcher

SimVars And Keys

When creating add-ons for aircraft - primarily Instruments - you will need to be able to access Simulation Variables and change them based on user input in the form of Key Events. Both using SimVar values and using input key events is done using operations written in RPN, as explained below. Note that you can also find many use-cases and examples in the different ModelBehavior templates.

SimVars

SimVars permit you to get the state of a component, or the state of something in the simulation, and they are generally accesed using the following RPN format:

```
(A:[simvar_name]:[simvar_index], [units])
```

Note that the "index" value is only required by a few SimVars. Here are some simple examples:

some simple examples:

```
(A:SIM ON GROUND, Bool)
(A:AIRSPEED TRUE, Knots)
(A:TURB ENG IGNITION SWITCH EX1:1, Enum)
```

Sometimes a SimVar will require **two** additional index values, for example the SimVar `CIRCUIT CONNECTION ON` requires a circuit index and a bus index. Normally this would be written like this:

```
2 (>A:BUS LOOKUP INDEX, Number) (A:CIRCUIT CONNECTION ON:45, Bool)
```

However, some SimVars will permit an alterative "shorthand" method of using the two indices, like this:

```
2 (A:1:CIRCUIT CONNECTION ON:45, Bool)
```

Here, the `1:` after the `A:` is used to tell the sim to send the top value on the stack to the SimVar. This shorthand method is ONLY applicable to the following SimVars:

- `BUS CONNECTION ON`
- `BATTERY CONNECTION ON`
- `ALTERNATOR CONNECTION ON`
- `CIRCUIT CONNECTION ON`
- `EXTERNAL POWER CONNECTION ON`
- `BUS BREAKER PULLED`
- `BATTERY BREAKER PULLED`
- `ALTERNATOR BREAKER PULLED`
- `CIRCUIT BREAKER PULLED`
- `EXTERNAL POWER BREAKER PULLED`

Finally it should be noted that sometimes you will see simulation variables written using *snake case*, and prefixed with `SIMVAR_`, as that is how they appear in the simulation code itself. However when using them in RPN, you should omit the `SIMVAR` part as well as the `_`. For example:

```
SIMVAR_CIRCUIT_ON -> (A:CIRCUIT ON:1, Bool)
```

Keys

Key events permit you to change the state of a component or other aspect of the simulation based on some form of input. When checking for a key event, you would use the following RPN format:

```
[0] (>K:[key_event_name])
[1] [0] (>K:2:[key_event_name])
```

There are two formats showing for accessing key events because many of them require not just one but *two* parameters, and so the formatting above shows how this is achieved. Below are two "real world" examples, one with a single parameter and the other with 2 parameters:

```
1 (>K:TOGGLE_EXTERNAL_POWER)
50 1 (>K:2:PANEL_LIGHTS_POWER_SETTING_SET)
```